

1 Simulate before you sample: a three-step loop for designing
2 ecological studies

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14 **Abstract**

15 1. Much of what an ecological study can reveal is fixed before any data are collected. Sample size,
16 treatment levels, the range of predictor values and measurement effort together determine which
17 effects can be estimated, and how precisely. Yet these choices are often made by convention,
18 or planned only against a significance threshold, which promotes dichotomous “significant / non-
19 significant” thinking from the outset.

20 2. Here we propose a practical addition to the research workflow: a *simulation loop*, carried out
21 entirely before data collection. The loop has three steps: (1) translate a conceptual hypothesis into
22 an explicit generative model; (2) simulate and visualise the data that model implies, to scrutinise
23 assumptions about effect sizes, variability and measurement limitations; and (3) iterate design

24 choices—sample size, treatment levels, predictor range, measurement effort—using criteria focused
25 on estimation quality, especially interval width (precision) and bias, rather than binary detect/not-
26 detect framing. We illustrate the loop with worked linear and non-linear examples and reproducible
27 R code, requiring no specialised statistical machinery beyond base R.

28 3. The examples show how the loop exposes identifiability limits that no increase in sample size can
29 resolve, how precision can depend as much on *where* you sample as on how much you sample, and
30 how design targets can be set from the precision a scientific or management claim actually requires.

31 4. Simulating before sampling does not guarantee correct inference; its value is conditional on the
32 assumptions explored. But making those assumptions explicit and inspectable becomes increas-
33 ingly important as ecological analyses draw on larger datasets and AI-assisted pattern discovery,
34 where deciding what to believe depends on clearly stated assumptions and defensible uncertainty
35 quantification. We intend the loop as a teachable entry point for researchers and students who have
36 never written a generative model.

37 **Keywords:** statistical methods, quantitative ecology, hypothesis testing, scientific reasoning, experi-
38 mental design, power analysis, study design, simulation, statistical power, precision, effect size, repro-
39 ducibility.

1 Introduction

Many of us have seen the same scene in a conference talk, an in-house seminar, or a paper. A speaker presents a compelling story, shows a clean figure with the mean response in a control and a treatment group, and points to the familiar stars of significance, concluding that the treatment “has an effect” ($p < 0.05$)—or, more problematically, that it “has no effect” ($p > 0.05$). The combination of a coherent narrative, an intuitive plot and a thresholded summary can make the conclusion feel settled, even though the same data are often compatible with a wide range of effect sizes. For many listeners and readers, this leaves a quieter set of questions: how much do we actually know from these data, what else are they compatible with, and would the interpretation change under a different but still reasonable analysis?

These concerns are not new. The binary, threshold-based style of statistical reasoning that many of us still use—and are often still taught to use—originated about a century ago (Neyman & Pearson 1933; Gigerenzer 2004) and has been criticised for much of that time, with especially clear calls in recent decades to move beyond “statistical significance” as a scientific standard (e.g. Anderson *et al.* 2000; Amrhein *et al.* 2017; Wasserstein *et al.* 2019; Amrhein *et al.* 2019). Yet the reform has been uneven in ecology and evolutionary biology. Surveys of the published literature find widespread under-powered designs, selective reporting and systematically exaggerated effect sizes (Kimmel *et al.* 2023; Yang *et al.* 2023; Fraser *et al.* 2018), and quantitative training often still centres on the very tests being criticised (Touchon & McCoy 2016).

Most of this reform effort has been directed at how results are analysed and reported. We think an equally important part of the problem sits earlier. By the time data exist, much of what a study can reveal has already been fixed by choices made months before: how many replicates, which treatment levels, how wide a gradient of conditions to span, and how carefully to measure. No subsequent modelling choice can recover information that the sampling design never collected. Worse, a design with poor precision does not simply fail to detect an effect. Conditional on passing a significance filter, such a design reports effects that are systematically too large, and sometimes of the wrong sign (Gelman & Carlin 2014). This is a plausible mechanism behind the exaggeration documented in the ecological literature (Kimmel *et al.* 2023), and it is a property of the design rather than of the analyst. The conventional tool for making these choices, power analysis, inherits the framing we are trying to escape. It asks whether a test will cross a threshold rather than how precisely a quantity will be known, and it is frequently misapplied after the data are in (Hoenig & Heisey 2001). Planning instead

71 for *precision*, that is, for an uncertainty interval narrow enough to support the claim at stake, has been
72 advocated for decades (Goodman & Berlin 1994; Cumming 2013; Rothman & Greenland 2018; Maxwell
73 *et al.* 2008; Lakens 2022). In ecology, however, precision-based planning meets a practical obstacle:
74 closed-form expressions exist only for the simplest designs, while the settings ecologists actually work
75 in, such as saturating responses, nested sampling and non-Gaussian data, have none. What is needed
76 is a way to reach the same answers by simulation.

77 Here we propose a small but practical addition to the research workflow: a *simulation loop*, carried out
78 before any data are collected. The loop has three steps. First, translate a conceptual hypothesis into
79 an explicit generative model, that is, a formula stating how the data would arise, including both the
80 expected response and its stochastic component. Second, simulate and visualise the data that model
81 implies, and ask whether they look like data the system could plausibly produce. Third, iterate the
82 design, that is, sample size, treatment levels, the range of predictor values and measurement effort,
83 judging each version by the precision it delivers for the parameter that matters rather than by whether
84 a test would clear a threshold. The procedure is a loop because each step can send us back to the
85 previous one: simulated data that look absurd send us back to the model, and a design that cannot
86 deliver useful precision sends us back to the question.

87 Simulation itself is not new in ecology and evolution, and our contribution is neither the idea nor the
88 machinery but a deliberately narrow and teachable version of them (see DiRenzo *et al.* 2023; Wolkovich
89 *et al.* 2026; Gelman *et al.* 2020; Schad *et al.* 2021; Morris *et al.* 2019, and Table 1 for how the loop
90 relates to other simulation-based approaches). We intend it as an entry point for researchers and
91 students who have never written a generative model. Throughout, we provide worked examples with
92 reproducible code: a linear case (Section 3), a non-linear case in which the design question sharpens
93 considerably (Section 4), and two ways in which a precise design can nonetheless mislead (Section 5).
94 Figure 9 places the loop within the conventional research workflow and marks the pitfalls it can help
95 anticipate; Box 1 defines those pitfalls and gives key references.

96 2 The simulation loop

97 This section sets out the loop in general terms; Sections 3 and 4 work through it on concrete examples.
98 The aim throughout is to make assumptions explicit, to turn a verbal hypothesis into a quantitative
99 expectation, and to find out what a planned design can and cannot support before it is carried out.

2.1 Core idea: an explicit generative model

At the centre of the loop is an explicit generative model, a statement of how the data would come about. A useful starting point separates the systematic component (pattern) from the stochastic component (noise):

$$y_i = f(x_i; \theta) + \varepsilon_i \quad (1)$$

where $f(x_i; \theta)$ is the expected value of y_i given predictors x_i and parameters θ , the signal implied by the hypothesis, and ε_i is the remaining deviation around that expectation. Depending on context, ε_i may represent process variation, observation or measurement error, or both. It is worth stating which, because the two have different design remedies: process variation is a property of the system and must be lived with, whereas measurement error can often be reduced by better instruments, repeated readings or a more careful protocol.

Equation 1 describes how y arises *given* x . For design purposes we must also specify how x itself arises: the treatment allocation, the range of predictor values, the replication structure, the spatial or temporal layout, the measurement schedule. This second half is easy to overlook, and it is where most of the design leverage lives. It is also what distinguishes a design simulation from a model check. The parameters θ are assumptions about nature, which we can only estimate or guess at; the process generating x is a description of what we intend to do, and is entirely under our control. Simulating both makes visible which features of the data are determined by our choices and which are left to chance.

We ask only that the model be explicit enough to simulate from. The more of it that rests on known mechanism, the better: mechanism is what constrains plausible parameter values, so Step 1 is easier and the resulting design more defensible when the model has one.

2.2 The three steps

Step 1: From a verbal hypothesis to a generative model. Models are hypotheses. With every model we fit we are testing an additional set of claims, often without full awareness. Building the model deliberately, from what is already known about the variables and their relationships, constrains this to a defensible set. Which explanatory variables influence y ? Is the relationship linear, saturating, or humped? What does the literature suggest about the magnitude of the effect, and about how variable

127 it is between individuals, plots or sites? In our experience a conceptual hypothesis often makes intuitive
128 sense right up to the moment one tries to write it as a formula, at which point it can become clear
129 that it lacks direction, magnitude, or any foundation in prior knowledge. This is uncomfortable, and
130 it is among the most valuable things the loop does. Writing the model down also forces us to state
131 what we actually want to estimate—the estimand—and what values we expect it to take. In ecology
132 this step should anticipate design features such as hierarchy and non-independence, since they change
133 what the data can support.

134 **Step 2: Simulate and visualise synthetic data.** “The greatest value of a picture is when it forces
135 us to notice what we never expected to see” (Tukey 1977). With the model specified, we generate a
136 synthetic dataset from it. The key point is to simulate y *through* the model structure rather than
137 drawing y from some marginal distribution: we generate predictor values under the planned design,
138 compute the expected response via $f(x; \theta)$, and then add stochasticity via ε . Plotting the result
139 shows the hypothesis in the same visual form as a real results figure, with direction, magnitude and
140 consistency together. The question to ask at this point is biological rather than statistical: could this
141 system plausibly produce these data? Implausible scatter, impossible values, or a relationship that
142 looks nothing like what one would expect from the literature are all signs that the assumptions, rather
143 than the simulation, need revisiting.

144 **Step 3: Iterate the design against a precision target.** The third step plays the planned study
145 through many times on the computer. This goes beyond a classical power analysis, which targets a
146 significance threshold. We repeatedly simulate data under plausible assumptions, fit the analysis we
147 intend to use, and examine the estimates and their uncertainty. Three questions follow, in order:

- 148 1. *Does the analysis recover the values we generated from?* This is a check on the code and on the
149 model implementation, not on the science. If a correct analysis cannot recover parameters from
150 data we generated ourselves under ideal conditions, something is wrong—and we would never
151 have learned it from the empirical fit alone.
- 152 2. *How do estimates and uncertainty respond to design choices?* Sample size, the range of predictor
153 values, the number of groups versus replicates per group, and measurement effort are all levers.
154 They differ enormously in cost, and their relative value is rarely obvious in advance.
- 155 3. *What would this design do to our conclusions if we interpreted it conventionally?* Simulation

156 makes it straightforward to compute quantities that are otherwise invisible: how often the esti-
157 mate would carry the wrong sign, and by what factor it would be exaggerated, conditional on
158 passing a significance filter (Gelman & Carlin 2014).

159 For this to end in a decision it needs a target. The translation is direct: if a claim requires an effect
160 to be known to within $\pm k$, the design target is an uncertainty interval of width about $2k$ (Rothman
161 & Greenland 2018). One refinement matters. A design does not produce a single interval width but a
162 distribution of widths across the studies one might run, so designing to the mean of that distribution
163 leaves roughly half of realised studies short of the target. Designing against a high quantile instead—
164 conventionally the 80th percentile—is known as *assurance* in the clinical trials literature and costs
165 nothing but a change of summary statistic. Section 3 works this through.

166 Going back and forth between assumptions, simulation and fitting is a playful but powerful way to
167 design a study. The procedure is a *loop* because each step can send us back to the previous one:
168 simulated data that look absurd send us back to the model, and a design that cannot deliver useful
169 precision sends us back to the question.

170 2.3 What the loop is, and what it is not

171 Simulation is used in several distinct ways in ecology and evolution, and the differences matter because
172 they answer different questions at different points in a project. Table 1 places the simulation loop
173 among them. Our scope is deliberately the narrowest of the five: everything in this paper happens
174 before any data exist, and the criterion throughout is the precision of the estimate we care about.
175 That narrowness has a cost, which Section 5 makes explicit: precision is a statement about sampling
176 variability alone, and says nothing about whether the model is right or whether the parameter is the
177 one the question was about.

178 3 A linear example: plant growth and nitrogen

179 The loop is best shown in practice. This first example is deliberately simple and can be reproduced
180 directly; a fully commented R script containing all the code and figures below is available in the
181 Supplement.

Table 1: Five uses of simulation in statistical practice. The simulation loop occupies the earliest and narrowest of these roles. It complements rather than replaces the others; a study designed with the loop would naturally be analysed with a workflow of the kind described by Wolkovich *et al.* (2026) or Gelman *et al.* (2020).

Approach	Question it answers	When applied	Criterion
Classical power analysis	Will my test reject the null hypothesis?	Before data	Power at a threshold
Simulation loop (this paper)	How precisely will I know the quantity I care about, and is that precise enough to be useful?	Before data	Interval width for the estimand
(Bayesian) statistical workflows (Gelman <i>et al.</i> 2020; Schad <i>et al.</i> 2021; Wolkovich <i>et al.</i> 2026)	Is my model, and my inference from it, trustworthy?	Before and during analysis	Calibration, parameter recovery
Validating complex models (DiRenzo <i>et al.</i> 2023)	Does this model behave as I think it does on data like mine?	After model specification	Recovery, identifiability
Simulation studies (Morris <i>et al.</i> 2019)	Does this statistical method perform well in general?	Method development	Bias, coverage, RMSE

3.1 Step 1: from a verbal hypothesis to a generative model

Every study starts with a question. Here: what is the effect of nitrogen fertilisation on plant growth? We plan to grow sunflowers across a gradient of nitrogen concentrations and harvest biomass after one season. The response y is biomass, and the verbal hypothesis is that plants grow more when nitrogen increases.

That sentence is not yet something we can simulate from. To simulate, we must commit to three things it leaves open: the *shape* of the relationship, the *size* of the effect, and its *consistency*. Assuming a positive linear relationship between biomass and nitrogen concentration x gives

$$y_i = a + b x_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma) \quad (2)$$

where a is the intercept, b is the slope—the expected biomass increase per unit nitrogen, and our estimand—and σ captures variability around the mean relationship. We now have to say what we think these parameters are:

```
# Our assumptions about the system, stated as numbers
a <- 30 # intercept: expected biomass (g) at zero nitrogen
b <- 7  # slope: biomass gain (g) per unit nitrogen; the estimand
sigma <- 5 # residual SD (g): process variation plus measurement error
```

These values are not truth. They are assumptions, made explicit so that they can be argued with, and everything that follows is conditional on them. That is the point rather than a limitation: it moves

201 the debate from the sample size to the reasoning behind it.

202 3.2 Step 2: simulate the data the model implies

203 We choose a design and generate data through the model structure: draw predictor values as the
204 design specifies, compute the expected response, then add noise.

```
205 # Choose a design and simulate one study from the model above
206 n <- 50 # number of plants
207 x <- rnorm(n, mean = 10, sd = 4) # planned nitrogen gradient
208 y <- a + b * x + rnorm(n, mean = 0, sd = sigma) # expected response, plus noise
209 fake <- data.frame(x = x, y = y)
210
```

212 In an experiment with fixed treatment levels rather than a gradient, only the line generating x changes,
213 for example `x <- rep(c(0, 5, 10, 15, 20), each = 10)`. The rest of the loop is identical.

214 Plotting the result (Fig. 1a) shows the hypothesis in the same visual form as a real results figure.
215 The question to ask is biological rather than statistical: could sunflowers in this system plausibly
216 produce these data? If the scatter looks far too tight, or the biomass values are impossible, then the
217 assumptions are wrong, and it is very much cheaper to discover that now than after a field season.

218 We can also fit the model we intend to use and check that it recovers what we put in (Fig. 1b):

```
219 # Fit the planned analysis and look at the interval, not the p-value
220 fit_lm <- lm(y ~ x, data = fake)
221 confint(fit_lm)["x", ] # 95% interval for the slope b
222
223
```

224 At this point it is tempting to look at the p -value. We deliberately do not. The quantity we carry
225 forward is the *width* of the interval for b (Fig. 1c). A narrow interval means the study pinned the
226 nitrogen effect down; a wide one means the data are compatible with many different effect sizes,
227 whatever the p -value says. Note also that this is a single study. Run it again and the interval shifts
228 and changes length.

Figure 1: Step 2 of the loop for the linear example. (a) Data simulated from Equation 2 with $a = 30$, $b = 7$, $\sigma = 5$ and $n = 50$: the hypothesis in visual form, showing effect size and scatter together. (b) The fitted model (solid) against the generating model (dashed). (c) The estimated slope and its 95% interval. The length of this bar is the quantity the rest of the section is about.

3.3 Step 3: how precision responds to design

We now play the planned study through many times. One run generates a dataset, fits the intended model, and records the interval width for b :

```
# One simulated study: generate data, fit the model, return the interval width
one_run <- function(n, a, b, sigma, mean_x = 10, sd_x = 4) {
  x <- rnorm(n, mean = mean_x, sd = sd_x)
  y <- a + b * x + rnorm(n, 0, sigma)
  ci <- confint(lm(y ~ x))["x", ]
  unname(ci[2] - ci[1])
}

# Repeat the same study many times to see what this design typically delivers
widths <- replicate(500, one_run(50, a, b, sigma))
```

Repeating this across sample sizes traces a design-precision curve (Fig. 2a). Precision improves with n , but with sharply diminishing returns: interval width falls roughly as $1/\sqrt{n}$, so halving the interval costs four times the sample size. The figure carries two lines because a design does not have a precision but a distribution of precisions (Fig. 2b), and the 80th percentile is the honest summary of what a single realised study is likely to achieve.

This particular case has a closed-form answer, $se(b) = \sigma / (s_x \sqrt{n-1})$, and the simulation reproduces it exactly (Fig. 2c). We show this deliberately: simulation was not necessary here, and a formula would have done. Building the machinery where it can be checked is what licenses using it in Sections 4 and 5, where no formula exists.

Sample size is not the only lever, and often not the best one. Holding $n = 50$ fixed and varying only the spread of nitrogen values shrinks the interval substantially (Fig. 2d):

```
# Same n, wider nitrogen gradient: how much does spreading x out buy us?
sd_x_values <- c(1, 2, 4, 6, 8)
sapply(sd_x_values, function(s)
  mean(replicate(500, one_run(50, a, b, sigma, sd_x = s))))
```

This lever is essentially free, since it costs no more to fertilise across a wider range, and it is invisible to a power calculation that treats n as the only design variable. Measurement quality is a third lever (Fig. 2e): doubling σ costs roughly a fourfold increase in n to recover the same precision, so a better instrument or a repeated reading may be far cheaper than more plants.

Figure 2: Step 3: how the precision of the nitrogen effect responds to design choices. (a) Design-precision curve: mean and 80th percentile of the 95% interval width against sample size. (b) At a single design ($n = 50$) the width is itself a random variable; the mean is a poor target because about half of realised studies fall beyond it. (c) The simulated curve reproduces the closed-form result, validating the machinery before it is used where no formula exists. (d) At fixed $n = 50$, spreading the predictor more widely improves precision markedly. (e) Doubling the residual SD costs roughly a fourfold increase in sample size.

3.4 From a scientific requirement to a sample size

Applying the rule from Section 2: suppose a fertiliser recommendation requires the nitrogen effect to within ± 0.25 g biomass per unit nitrogen. The design target is then an interval of width 0.5, read off the curve at the 80th percentile (Fig. 3).

```
# What sample size meets a target width of 0.5 in 80% of studies?
n_grid <- seq(20, 300, by = 20)
q80    <- sapply(n_grid, function(nn)
  quantile(replicate(500, one_run(nn, a, b, sigma)), 0.8))
n_grid[which(q80 <= 0.5)[1]]
```

Under the assumptions above this gives $n \approx 120$. The number is conditional on the assumed σ and predictor range, which is a feature rather than a limitation: it makes the basis of the design auditable. A reviewer who disagrees can argue with σ , a statement about the system, rather than with the sample size, which is merely its consequence.

Figure 3: Choosing a sample size from a precision requirement. The curve shows the 80th percentile of the 95% interval width for b ; the horizontal line is the target width of 0.5 implied by a requirement of ± 0.25 . The two meet at $n \approx 120$.

3.5 What an imprecise design would do to our conclusions

Everything so far concerns precision. The same machinery also reveals what poor precision does to *interpretation*, which is where the consequences for the published literature lie.

Consider a more typical ecological situation than our flagship example: a small true effect ($b = 0.5$), noisy measurement ($\sigma = 15$) and a modest sample ($n = 20$). Simulating this design and applying a conventional significance filter gives a power of about 9%. That much a power analysis would have told us. What it would not have told us is the rest. Conditional on reaching $p < 0.05$, this design reports an effect roughly four times too large, and carries the wrong sign about 7% of the time (Gelman & Carlin 2014). The distribution of estimates that would be published barely overlaps the truth (Fig. 4).

289 The design therefore does not merely fail to detect the effect; it manufactures misleading results. This
 290 cannot be diagnosed after the fact from a single dataset, because the exaggeration is invisible from
 291 inside the study that produced it. It is, however, entirely visible beforehand, from the design alone.

Figure 4: The distribution of slope estimates from 4000 simulated studies with $b = 0.5$, $\sigma = 15$ and $n = 20$ (grey), and the subset that would reach $p < 0.05$ and so be most likely to be reported (dark). Conditional on significance, the design exaggerates the effect roughly fourfold and returns the wrong sign about 7% of the time. The dashed line is the true value.

292 4 A non-linear example: a saturating response to nitrogen

293 Many ecological relationships are non-linear: responses saturate, show thresholds, or peak at interme-
 294 diate conditions. Simulation earns its place here, for two reasons. There is no closed-form expression
 295 for the precision of a non-linear parameter, so the analytic shortcut used in Section 3 is unavailable.
 296 And the same data can often be explained by quite different parameter combinations, which makes
 297 what you can learn depend heavily on where you sample rather than simply on how much.

298 4.1 Step 1: a saturating model

299 We keep the same question, but now assume that growth increases with nitrogen and then levels off
 300 as other resources become limiting. A standard choice is the Michaelis–Menten function:

$$y_i = \frac{V_{\max} x_i}{K + x_i} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma) \quad (3)$$

301 where $V_{\max}$ is the asymptotic biomass, the value approached as nitrogen becomes unlimiting, and K
 302 is the half-saturation constant, the nitrogen value at which expected biomass is $V_{\max}/2$.
 303 Unlike the slope in Section 3, these two parameters answer different biological questions. $V_{\max}$ describes
 304 the ceiling; K describes how quickly that ceiling is approached. A study can pin down one without
 305 the other, or neither, and which of those happens is largely a matter of design.

```
306 # Assumed parameter values for the saturating model
307 Vmax <- 120 # asymptotic biomass (g)
308 K <- 8 # half-saturation constant: nitrogen at which y = Vmax / 2
309 sigma <- 10 # residual SD (g)
310
311 # The mean function
312 mu_mm <- function(x, Vmax, K) Vmax * x / (K + x)
313
```

Equation 3 forces $y = 0$ at $x = 0$, which is unrealistic if the soil already contains nitrogen. Adding a baseline term is straightforward and makes the identifiability problem below somewhat worse rather than better; we keep the simpler form for clarity.

4.2 Step 2: two candidate designs, same sample size

The key design choice in a non-linear model is the range of x . We simulate two designs with identical sample sizes: one sampling nitrogen only in a low to moderate range, and one extending far enough to approach the asymptote.

```
# Two designs, same n, differing only in how far the nitrogen gradient extends
n <- 50
x_narrow <- runif(n, 0, 12) # Design A: stops well short of saturation
x_wide <- runif(n, 0, 40) # Design B: extends past the bend

fake_narrow <- data.frame(x = x_narrow,
  y = mu_mm(x_narrow, Vmax, K) + rnorm(n, 0, sigma))
fake_wide <- data.frame(x = x_wide,
  y = mu_mm(x_wide, Vmax, K) + rnorm(n, 0, sigma))
```

Plotting the two datasets shows the problem immediately (Fig. 5). Over the narrow range the saturating curve is nearly a straight line. The data contain almost no information about the asymptote, because the design never went anywhere near it. Same n , same model, same noise, but Design A cannot see the thing the model is about.

Figure 5: Two designs with the same sample size ($n = 50$) under Equation 3 with $V_{\max} = 120$, $K = 8$ and $\sigma = 10$. (a) A narrow nitrogen range gives data that look almost linear and carry little information about the asymptote. (b) A wider range reveals the curvature and the onset of saturation. The solid line is the generating model in both panels.

4.3 Step 3: fitting, convergence, and the cloud of estimates

We fit the planned model with non-linear least squares. Because `nls` can fail to converge, it is tempting to set `warnOnly = TRUE` and move on. We recommend against it: that option returns non-converged fits as though they were estimates, and quietly contaminates everything computed downstream. Recording convergence status instead turns a nuisance into a design diagnostic.

```
# Fit the saturating model; return NULL rather than a non-converged fit
fit_mm <- function(dat) {
  fit <- try(nls(y ~ Vmax * x / (K + x), data = dat,
```

```

346       start = list(Vmax = max(dat$y), K = median(dat$x)),
347       control = nls.control(maxiter = 200), silent = TRUE)
348       if (inherits(fit, "try-error")) NULL else fit
349   }
350

```

351 Under Design A the fit fails to converge in XX% of simulated studies, against YY% under Design B.
352 That difference is a result in its own right: it says the narrow design is asking the data for something
353 they cannot supply. A caution follows immediately. Discarding failed runs before summarising the rest
354 is itself a selection step, of the kind this paper warns against elsewhere, since the surviving estimates
355 are the subset that happened to be informative. How many runs were dropped should be reported
356 alongside the estimates that remain.

357 Repeating the whole study many times under each design and plotting the resulting parameter es-
358 timates against one another makes the underlying problem visible (Fig. 6). Under Design A the
359 estimates smear along a ridge: many combinations of K and $V_{\max}$ produce almost identical curves
360 over $x \in [0, 12]$, so the data cannot tell them apart. A large $V_{\max}$ paired with a large K looks much
361 the same, over that range, as a smaller $V_{\max}$ paired with a smaller K . Under Design B the cloud col-
362 lapses around the true values. This is a design problem rather than an analysis problem: no cleverer
363 statistical method recovers information that the sampling design never collected.

Figure 6: Estimates of K and $V_{\max}$ from 500 simulated studies under each design, with the true values marked.
(a) Under the narrow design the two parameters trade off against one another and neither is well determined.
(b) Under the wide design both are recovered. Non-converged fits are excluded; their frequency is reported in
the text.

364 4.4 More data, or better placed data?

365 The design question can now be put directly. Given a fixed budget, is it better to collect more
366 observations or to spread them over a wider range of nitrogen?

```

367   # Compare the spread of K estimates across sample sizes and nitrogen ranges
368   ns <- c(20, 50, 100, 200)
369   sapply(ns, function(nn) IQR(replicate_K(nn, x_max = 12))) # narrow
370   sapply(ns, function(nn) IQR(replicate_K(nn, x_max = 40))) # wide
371

```

373 Quadrupling the sample size under the narrow design buys less than simply extending the nitrogen
374 gradient at the original sample size (Fig. 7). If a project can afford one change, the simulation says
375 which one to make, and it says so before the money is spent. This is the central practical claim of the

376 paper in its sharpest form: identifiability and precision can depend as much on where you sample as
377 on how many observations you collect.

Figure 7: Interquartile range of estimates of K against sample size, under the narrow and wide nitrogen ranges. Extending the range at $n = 50$ achieves more than quadrupling the sample size within the narrow range.

378 So far the analysis model has always matched the model that generated the data, and the design has
379 always been experimental. Section 5 relaxes each assumption in turn, and shows that neither failure
380 is visible in the precision of the result.

381 5 When precision is not enough

382 Sections 3 and 4 treated precision as the design criterion. That is deliberate, but it is not sufficient,
383 and this section sets the bound. A 95% interval is a statement about sampling variability under an
384 assumed model. It says nothing about whether that model has the right structure, and nothing about
385 whether the parameter it surrounds is the one the scientific question was about. We demonstrate each
386 failure in turn. In both, the interval is narrow, the diagnostics are clean, and the answer is wrong.

387 5.1 A model with the wrong shape

388 We fit an ordinary straight line to data generated by the saturating process of Section 4, using the
389 narrow design in which the curve looks almost linear anyway.

```
390 # Fit a straight line to data generated by the saturating model
391 fit_wrong <- lm(y ~ x, data = fake_narrow)
392 summary(fit_wrong)$r.squared           # looks like a good fit
393 confint(fit_wrong)["x", ]              # a tight interval
394 predict(fit_wrong, newdata = data.frame(x = 40)) # extrapolate to x = 40
395 mu_mm(40, Vmax, K)                    # what the truth would be
396
397
```

398 Every conventional diagnostic looks reassuring. The model explains about three-quarters of the vari-
399 ance, the slope is estimated to within a couple of per cent, and nothing in the output flags a problem.
400 The predicted biomass at $x = 40$ is about 234 g. The true value is 100 g (Fig. 8a). Collecting more
401 data would tighten that interval without moving it, because the interval tightens around a value that
402 was never the target.

There is a constructive reading. The same simulation that exposes the problem points to the remedy: a design extending past the bend both identifies the parameters (Section 4) and makes the inadequacy of a linear model obvious in the data. Designing for precision and designing to be able to detect one's own mistakes turn out, here, to be the same activity.

5.2 A parameter that is not the one we mean

Now suppose the model has the right shape but we cannot randomise. We survey plots across a landscape, measuring nitrogen where it happens to occur. Soil moisture varies across that landscape and affects both: wetter plots mineralise more nitrogen, and wetter plots also grow more biomass for reasons unrelated to nitrogen.

The step that matters is Step 1, and it is easy to skip. The generative model must describe how we believe the data actually arise, including processes we do not intend to measure. Writing moisture into the simulation is not a claim that we will have moisture data; it is a claim about the world.

```
# An observational design in which moisture drives both nitrogen and biomass
n_obs    <- 500
moisture <- rnorm(n_obs, 0, 1)                # the confounder
x_obs    <- 10 + 3 * moisture + rnorm(n_obs, 0, 1)  # N tracks moisture
y_obs    <- 30 + 7 * x_obs + 15 * moisture + rnorm(n_obs, 0, 5)

naive <- lm(y_obs ~ x_obs)                    # moisture unmeasured
adj   <- lm(y_obs ~ x_obs + moisture)         # moisture measured
```

The causal effect of nitrogen is 7. The naive analysis returns about 11.5, with a 95% interval of roughly [11.3, 11.7] that excludes the truth entirely; the adjusted analysis recovers 6.9 (Fig. 8b). Nothing in the naive output signals a problem: the residuals are well behaved, the model is correctly specified as a straight line, and the estimate is very precise.

Precision is again the difficulty. The bias does not depend on n , so a larger sample narrows the interval without correcting the estimate. At $n = 10\,000$ the naive interval spans 11.44 to 11.53 (Fig. 8c). A large dataset, analysed with a correctly specified model, yields an extremely confident answer to a question nobody asked.

5.3 What follows for the loop

The simulation loop does not solve either problem. It cannot tell you that your functional form is wrong, and no amount of simulation licenses a causal claim from an observational design; that requires

Figure 8: Two ways a precise design can mislead. (a) A straight line fitted to data generated by the saturating model over a narrow nitrogen range (shaded): the fit is good by conventional criteria, yet extrapolating to $x = 40$ overpredicts biomass by more than a factor of two. (b) Estimates of the nitrogen effect at $n = 500$ from an analysis omitting soil moisture (naive) and one including it (adjusted); the dashed line is the true causal effect of 7. (c) The naive estimate as sample size increases from 100 to 10 000: the interval contracts steadily around a value roughly 65% too large.

randomisation or an explicit argument about which variables must be measured and adjusted for (Pearl 2019; Grace & Irvine 2020; Arif & MacNeil 2023).

What it offers is earlier warning, and three things follow. It quantifies the damage while it can still be acted on: simulating a confounder one suspects exists takes a few lines and answers a concrete question, and here the answer is 65%, large enough to change what one would do. It disciplines the estimand, because the naive analysis above is a perfectly good estimate of an association and wrong only as an answer to a causal question, a distinction that writing the generative model forces into the open. And it corrects a natural intuition about sample size: it is tempting to treat a large dataset as insurance against error, whereas for this class of problem the larger study is the more confidently wrong one, and the less likely to be questioned.

This is why we treat Step 1 as the step that repays most attention, and the one most often skipped. The precision machinery of Sections 3 and 4 is only as good as the generative model it is conditioned on.

6 Reporting a simulation loop

The loop is only useful to a reader if they can disagree with it productively. That requires reporting the assumptions, not just the resulting design. The checklist below lists what we think a paper or proposal should state; all of it fits comfortably in a short methods paragraph plus a supplement.

The same three steps extend without modification to two-group comparisons, generalised responses such as counts and proportions, and hierarchical or repeated-measures designs; only the generative model and the analysis model change. The Supplement works through a nested design in which a fixed budget of plots is allocated across sites in five different ways, a case where no closed-form guidance exists and the differences in precision are substantial.

1. **The generative model.** Equations for the mean function $f(x; \theta)$ and for the stochastic component ε , on the response scale.
2. **The assumed parameter values,** with their justification: literature, pilot data, or expert judgement labelled as such. State the range explored where a single value could not be defended.
3. **What the error term represents:** process variation, measurement error, or both, and whether hierarchy or autocorrelation is included.
4. **The design variables explored:** sample size, treatment levels, predictor range, replication structure, measurement effort. Say which were varied and which were held fixed.
5. **The performance criterion,** and the target. We recommend the width of the uncertainty interval for the focal parameter, summarised at a stated quantile rather than as a mean, together with the scientific or management requirement the target derives from.
6. **Failures as well as successes:** the proportion of simulated fits that did not converge, and how they were handled. This is a design diagnostic, not a nuisance.
7. **Reproducibility:** the code, the random seed, and software versions, archived with a persistent identifier.

7 Outlook

The proposal of this paper is narrow. Turn a verbal hypothesis into an explicit generative model, simulate what it implies, and choose a design by the precision it delivers for the parameter that matters. Everything else follows from taking that seriously, including the uncomfortable findings of Section 5.

Advice of this kind is not new, and it is worth asking why it has not been taken up more widely. We think the honest answer is that it was expensive. Writing a generative model for anything beyond a straight line has required a level of comfort with code that ecological training has mostly reserved for theoreticians, and an empiricist facing a field season has had good reason to skip it. That barrier is

462 now falling quickly. Code assistants make a first simulation of a nested, non-Gaussian or saturating
 463 design something that can be drafted in minutes rather than an afternoon, and revised as easily. The
 464 practical case for the loop is therefore stronger today than when much of the same advice was offered a
 465 decade ago, not because the argument has improved but because the cost of acting on it has dropped.
 466 What remains expensive is the part that matters most. A model that can be simulated from still has
 467 to be defended: which processes generate the data, what magnitudes are plausible, what varies and by
 468 how much, and what the study is actually trying to estimate. Those judgements are ecological rather
 469 than computational, and nothing about cheaper code supplies them. If anything the balance shifts
 470 further towards Step 1, because the steps that used to consume the effort no longer do. The same is
 471 true of the design target itself. Choosing an interval width of $2k$ is a statement about how much error
 472 is tolerable for the claim at stake, which is a scientific and sometimes a policy judgement rather than
 473 a statistical one (Douglas 2000; Steel 2010). Making the generative model explicit does not settle such
 474 questions, but it turns them from hidden defaults into inspectable choices (Romeijn 2014; Gelman &
 475 Shalizi 2013), which is the most that can be asked of any method (Box 1976).
 476 These considerations become more pressing, not less, as ecological analyses draw on larger datasets
 477 and on AI-assisted pattern discovery. Modern methods are very good at finding regularities in high-
 478 dimensional data and comparatively poor at telling us which of them to believe (Pichler & Hartig
 479 2023; Lucas 2020). Simulation offers one concrete form of discipline here: a generative model specifies
 480 what the data would look like if a proposed mechanism were absent, and a pattern that a method also
 481 recovers from such data has not been established by finding it. The confounded example of Section 5 is
 482 the more sobering case, because it scales in the wrong direction. Larger samples narrowed the interval
 483 without moving the estimate, so the studies that command most confidence are the ones whose bias is
 484 hardest to notice. Volume of data is not a substitute for having asked what generated it (Pearl 2019),
 485 and reporting standards for uncertainty in ecology remain uneven enough that the difference is easy
 486 to miss (Simmonds *et al.* 2024; Lele 2020).
 487 None of this requires a new methodology. It requires doing something ordinary at an unusual time,
 488 namely writing down what one expects before finding out, and it costs an afternoon at the point in
 489 a project where an afternoon is cheapest. We have found the exercise most valuable when it went
 490 badly: when the simulated data looked nothing like the system, when the estimates would not settle
 491 however much data we imagined collecting, or when the design we had planned turned out to answer a
 492 question adjacent to the one we cared about. Those are useful things to learn in front of a computer,

⁴⁹³ and expensive things to learn in a field.

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Box 1: Nine pitfalls along the research workflow

Numbered as in Figure 9, in the order they arise along the workflow. • marks pitfalls a design choice can fix, and which the simulation loop therefore addresses directly; ◦ those it can expose or quantify but not resolve.

Framing the question and hypothesis

◦ **(1) Estimating something other than the quantity of interest.** An association is reported as though it were a causal effect, because a driver of both the predictor and the response was unmeasured, or because sampling was selective. The estimate may be perfectly good as a description and still answer a different question from the one asked. Refs: Pearl 2019; Grace & Irvine 2020; Arif & MacNeil 2023.

◦ **(2) Decisions made after seeing the data.** Hypotheses adjusted to fit results, outcomes or subsets chosen once their behaviour is known, and analyses continued until they look right. This rarely feels like misconduct, because each individual choice is defensible; what is not defensible is that the choices were available. Refs: Gelman & Loken 2013; Kerr 1998; Stefan & Schönbrodt 2023; Nickerson 1998.

• **(3) A hypothesis that cannot be wrong.** The claim has no stated direction, no expected magnitude, and no defined alternative, so almost any result can be read as support. Vagueness survives argument but not arithmetic: a hypothesis that cannot be written as a formula with numbers in it was never specific enough to test. Refs: Rajtmajer *et al.* 2022; Gigerenzer 2004; Perezgonzalez 2015.

Designing the study

• **(4) Designing to detect rather than to estimate.** Sample size chosen so that a test is likely to cross a significance threshold, rather than so that the quantity of interest will be known precisely enough to support the claim at stake. The two questions have different answers, and the second is the one a reader needs. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Hoenig & Heisey 2001; Halsey *et al.* 2015.

• **(5) Sampling that cannot identify the target.** Predictor values spanning too narrow a range, treatment levels placed where the response is flat, or too few groups to estimate a group-varying effect. Where the observations are taken can matter more than how many there are, and no increase in sample size compensates for a design that never visits the conditions the parameter

describes. Refs: DiRenzo *et al.* 2023; this paper, Section 4.

- **(6) Ignoring how the data actually arise.** Measurement error, imperfect detection, misclassification, and observations that are not independent because they share a plot, a site, or a time. Unlike variation in the system itself, much of this is a design choice: better instruments, repeated readings, or a different allocation of effort. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

Analysing and reporting

- **(7) A model that does not match the process.** A saturating response fitted as a straight line, the wrong distribution or link, omitted grouping structure, a transformation that changes what the parameters mean, or extrapolation beyond the range sampled. Such a model can fit well, report a narrow interval, and still be badly wrong about the quantity that matters; regression to the mean is a familiar special case. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024; Barnett *et al.* 2005; Wolkovich *et al.* 2021.

- **(8) Conclusions that turn on a threshold.** Results summarised as present or absent rather than as a magnitude with uncertainty, and non-significant findings read as evidence of no effect when they are usually compatible with effects large enough to matter. Reporting the interval instead removes the problem at source. Refs: Amrhein *et al.* 2017, 2019; Wasserstein *et al.* 2019; Muff *et al.* 2022; Singal *et al.* 2017.

- **(9) The significance filter on the literature.** Because significant results are more likely to be written up and published, the published subset of an imprecise design overstates the effect and sometimes reverses its sign, even when every individual study was analysed correctly. The exaggeration is invisible from inside any one study but computable from the design in advance. Refs: Gelman & Carlin 2014; van Zwet & Cator 2021; Kimmel *et al.* 2023; Yang *et al.* 2023; Ioannidis 2005.

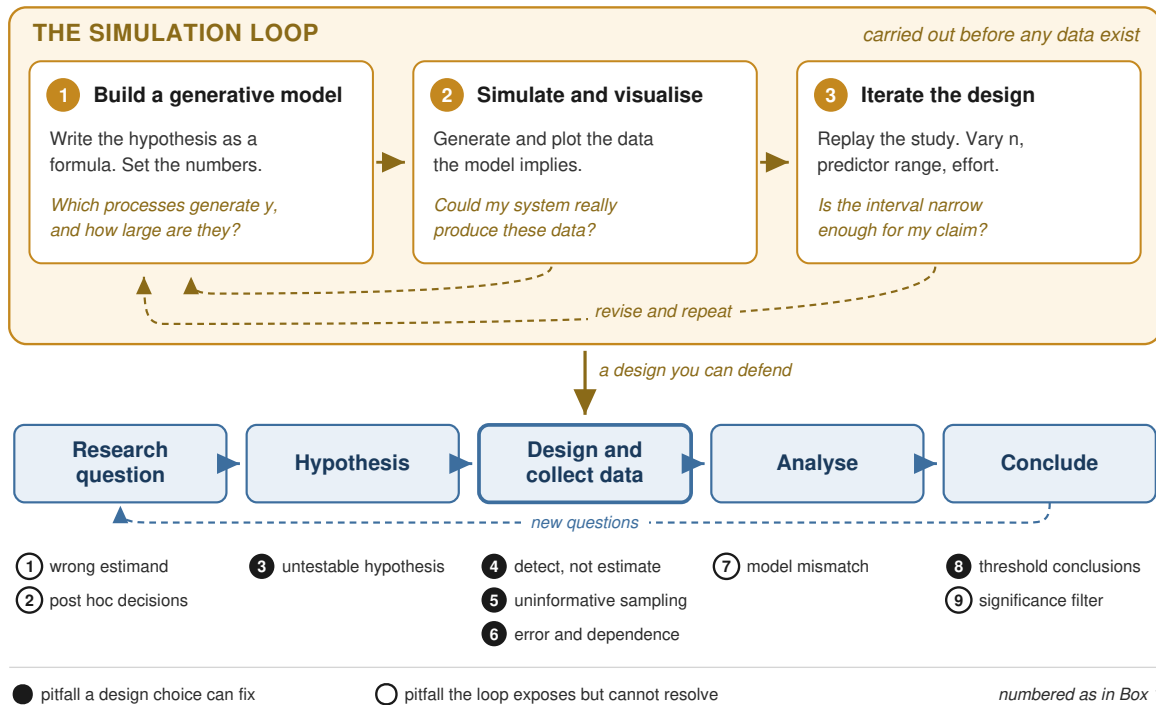


Figure 9: The simulation loop within the conventional research workflow. The lower row shows the familiar sequence from research question to conclusion, with a dashed return path indicating that conclusions raise new questions. The simulation loop (upper panel) is inserted between forming a hypothesis and collecting data, and runs entirely before any observation is made. Its three steps are to translate the hypothesis into an explicit generative model, to simulate and plot the data that model implies, and to iterate the design against a target precision for the parameter of interest. Arrows within the loop run in both directions: simulated data that look implausible send one back to the model, and a design that cannot deliver useful precision sends one back to the question. Numbered markers show the stage at which each pitfall of Box 1 arises. Filled markers (●) are pitfalls a design choice can fix, and which the loop therefore addresses directly; open markers (○) are those it can expose or quantify but not resolve. The loop is placed where it is because sample size, treatment levels, predictor range and measurement effort together fix what a study can reveal, and no later choice of analysis recovers information the design did not collect.